

Figure J-2 A V-bottom potential.

contains all possible information concerning the application of a perturbation $v(x)$ to a system whose unperturbed eigenfunctions are $\psi_1(x)$, $\psi_2(x)$, $\psi_3(x)$, $\psi_4(x)$, $\dots$

AN EXAMPLE

Let us illustrate the use of (J-13a) and (J-14) by doing a simple perturbation calculation. We shall evaluate the first eigenvalue and eigenfunction for the potential indicated in Figure J-2 and specified by the equation

$$V(x) = \begin{cases} \delta \frac{|x|}{a/2} & -a/2 < x < +a/2 \\ \infty & x < -a/2 \text{ or } x > +a/2 \end{cases} \quad (\text{J-16})$$

We consider this as the sum of an unperturbed potential

$$V(x) = \begin{cases} 0 & -a/2 < x < +a/2 \\ \infty & x < -a/2 \text{ or } x > +a/2 \end{cases}$$

which is an infinite square well, plus a perturbation

$$v(x) = \delta \frac{|x|}{a/2}$$

According to (6-79), (6-80), and Example 5-10, the normalized unperturbed eigenfunctions can be written

$$\psi_m(x) = \begin{cases} \sqrt{2/a} \cos(m\pi x/a) & m = 1, 3, 5, \dots \\ \sqrt{2/a} \sin(m\pi x/a) & m = 2, 4, 6, \dots \end{cases}$$

According to (6-81), the unperturbed eigenvalues are

$$E_m = \pi^2 \hbar^2 m^2 / 2Ma^2 \quad m = 1, 2, 3, 4, \dots$$

where we use M for the mass of the particle. If δ is small compared to the first eigenvalue

$$E_1 = \pi^2 \hbar^2 / 2Ma^2$$

the perturbation technique should be applicable.

To evaluate $\psi'_1(x)$, take $n = 1$ in (J-14). This gives

$$a_{1m} = \frac{1}{E_1 - E_m} \int_{-\infty}^{\infty} \psi_m^*(x) v(x) \psi_1(x) dx \quad m \neq 1$$

which is

$$a_{1m} = \begin{cases} \frac{8M\delta}{\pi^2 \hbar^2 (1 - m^2)} \int_{-a/2}^{a/2} \cos\left(\frac{m\pi x}{a}\right) |x| \cos\left(\frac{\pi x}{a}\right) dx & m = 3, 5, 7, \dots \\ \frac{8M\delta}{\pi^2 \hbar^2 (1 - m^2)} \int_{-a/2}^{a/2} \sin\left(\frac{m\pi x}{a}\right) |x| \cos\left(\frac{\pi x}{a}\right) dx & m = 2, 4, 6, \dots \end{cases}$$

For $m = 2, 4, 6, \dots$ the integrand is an odd function of x . Since the integral is taken over a range symmetrical about $x = 0$, the integral will vanish. Thus we have

$$a_{1m} = 0 \quad m = 2, 4, 6, \dots$$

For $m = 3, 5, 7, \dots$ the integral is an even function of x ; it gives

$$a_{1m} = \frac{16 M \delta}{\pi^2 \hbar^2} \frac{1}{(1-m^2)} \int_0^{a/2} \cos\left(\frac{m\pi x}{a}\right) x \cos\left(\frac{\pi x}{a}\right) dx \quad m = 3, 5, 7, \dots$$

Let $Z = \pi x/a$; then this becomes

$$a_{1m} = \frac{8}{\pi^2} \frac{\delta}{E_1} \frac{1}{(1-m^2)} \int_0^{\pi/2} \cos(mZ) Z \cos Z dZ$$

where we have introduced the convenient dimensionless ratio $\delta/E_1 = 2Ma^2\delta/\pi^2\hbar^2$. The integral can be evaluated easily by writing $\cos(mZ) = (1/2)(e^{+imZ} + e^{-imZ})$. The result is

$$a_{1m} = \frac{8}{\pi^2} \frac{\delta}{E_1} \frac{1}{(1-m^2)} \left\{ \frac{\cos[(m+1)\pi/2] - 1}{2(m+1)^2} + \frac{\cos[(m-1)\pi/2] - 1}{2(m-1)^2} \right\} \quad m = 3, 5, 7, \dots$$

The first few non-vanishing coefficients have the values

$$\begin{aligned} a_{13} &= \frac{1}{32} \frac{8}{\pi^2} \frac{\delta}{E_1} \\ a_{15} &= \frac{1}{864} \frac{8}{\pi^2} \frac{\delta}{E_1} \\ a_{17} &= \frac{1}{1728} \frac{8}{\pi^2} \frac{\delta}{E_1} \\ a_{19} &= \frac{1}{8000} \frac{8}{\pi^2} \frac{\delta}{E_1} \end{aligned}$$

It is not surprising that $a_{1m} = 0$ for $m = 2, 4, 6, \dots$. The perturbed potential $V'(x)$ is symmetrical about the origin, and so its first eigenstate must be of even parity. Consequently there can be no odd parity unperturbed eigenfunctions mixed into the first perturbed eigenfunction, and the odd parity unperturbed eigenfunctions are precisely those for $m = 2, 4, 6, \dots$. The perturbed eigenfunction $\psi'_1(x)$ is obtained by substituting the a_{1m} in the series (J-15). Since the a_{1m} decrease rapidly with increasing m (owing partly to the $1/(E_1 - E_m)$ term and partly to the v_{m1} term), it is apparent that we can get a very good approximation to the series by taking only the terms for $m = 1$ and $m = 3$. Thus

$$\psi'_1(x) \simeq a_{11}\psi_1(x) + \frac{1}{32} \frac{8}{\pi^2} \frac{\delta}{E_1} \psi_3(x) \quad (\text{J-17})$$

Finally the coefficient a_{11} must be adjusted so that $\psi'_1(x)$ is normalized, but we leave this as an exercise for the student.

Figure J-3 illustrates (J-17). The relative amount of $\psi_3(x)$ has been exaggerated for the sake of clarity. Fixing our attention on $\psi'_1(x)$ and $\psi_1(x)$, we see that the second derivative of the perturbed eigenfunction is relatively small near the ends of the region $-a/2$ to $+a/2$, and relatively large near the center, compared to the second derivative of the unperturbed

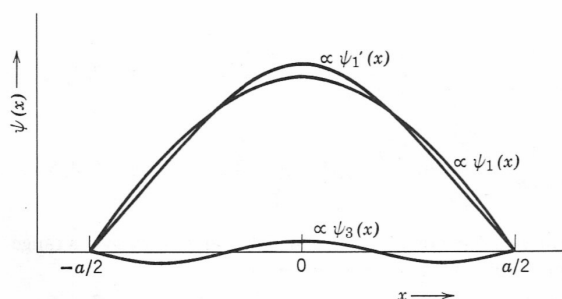


Figure J-3 Illustrating the composition of the first eigenfunction for a V-bottom potential.

eigenfunction. Consideration of the form of the time-independent Schroedinger equation for the perturbed and unperturbed potentials will make it clear why this happens.

Next let us evaluate $E'_1 - E_1$. Taking $n = 1$ in (J-13a), and inserting the appropriate unperturbed eigenfunction, we have

$$E'_1 - E_1 = \frac{4\delta}{a^2} \int_{-a/2}^{a/2} |x| \cos^2 \left(\frac{\pi x}{a} \right) dx$$

$$E'_1 - E_1 = \frac{8\delta}{a^2} \int_0^{a/2} x \cos^2 \left(\frac{\pi x}{a} \right) dx$$

$$E'_1 - E_1 = \frac{8\delta}{\pi^2} \int_0^{\pi/2} Z \cos^2 Z dZ$$

$$E'_1 - E_1 = \frac{8\delta}{\pi^2} \left(\frac{\pi^2}{16} - \frac{1}{4} \right)$$

which is

$$E'_1 - E_1 = 0.297\delta$$

Figure J-4 shows the perturbed eigenvalue E'_1 in terms of the dimensionless ratio $(E'_1 - E_1)/E_1$, plotted as a function of the dimensionless ratio δ/E_1 . Perturbation theory predicts the straight line of slope 0.297. The points are the correct answer. They were calculated from the eigenvalues E'_1 obtained by an accurate (numerical integration) solution of the time-independent Schroedinger equation for the four potentials $V'(x)$ corresponding to the values of δ/E_1 indicated. The shift in the energy of the first eigenvalue, as predicted by perturbation theory, is seen to be in error by about 10 percent for $\delta/E_1 \approx 0.9$, which corresponds to $(E'_1 - E_1)/E_1 \approx 0.25$. For $(E'_1 - E_1)/E_1 \approx 0.05$, the error is about 0.5 percent. Now it is apparent that the error in the perturbation theory we have developed is of the order of the square of a small quantity since, throughout the development, the squares of small quantities were always neglected. The numbers just quoted indicate that, in the present case, an approximate measure of the size of this small quantity is the ratio $(E'_1 - E_1)/E_1$. Note also that the eigenvalue E'_1 calculated by perturbation theory is always too large. It can be shown that this is true for any form of the perturbation $v(x)$, and it is easy to see why it happens. Perturbation theory uses the unperturbed eigenvalue $\psi_1(x)$ to evaluate $E'_1 - E_1 = \int_{-\infty}^{\infty} \psi_1^*(x) v(x) \psi_1(x) dx$. Comparing the plots of $\psi_1(x)$ and of $\psi'_1(x)$, we see that this procedure gives too much weight to the values of $v(x)$ near the ends of the region. But near the ends of the region $v(x)$ is largest, and therefore the contribution of $v(x)$ to the perturbed eigenvalue E'_1 is overestimated.

A comparison of the exact form of the eigenfunction $\psi'_1(x)$ of the potential $V'(x)$ with the form (J-17) predicted by perturbation theory shows that the error in the coefficient a_{13} is also of the order of the square of the quantity $(E'_1 - E_1)/E_1$.

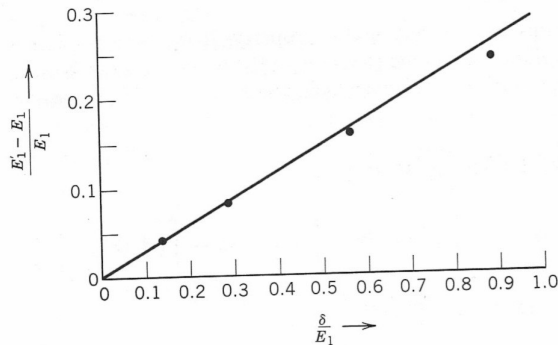


Figure J-4 A comparison between the first eigenvalues for several V-bottom potentials obtained from time-independent perturbation theory and from accurate solutions of the time-independent Schroedinger equation.